QUENTIN MORAINE

Robotics Engineering Student | MSc Robotics, Cranfield University
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PROFILE

French engineering student on a double degree with ECAM Arts et Métiers and the MSc Robotics at Cranfield. Has trained a vision model for a sorting machine, simulated an ABB robot cell in RobotStudio, built the forward and inverse kinematics of a UR3 arm in MATLAB and worked out the design basis of a cooling system for a district-heating plant. **Looking for a 6 to 7-month internship in robotics, automation or embedded systems, starting May 2027.**

PROJECTS

Vision-based screw-sorting machine – ECAM, project lead of a team of 4 **Sep 2025 – May 2026**

- Trained a YOLO11n model to recognise four screw head types, reaching 0.979 mAP50 on the validation set.
- Measured screw length and diameter with OpenCV and drove the sorting actuators from a Raspberry Pi 5 with a Flask interface.

ABB IRB 120 drawing cell – Cranfield, ABB RobotStudio **Oct 2026**

- Generated a path of about 220 targets from a CAD logo, then checked reach, robot configurations and approach moves in simulation before handing in the station.

UR3 and UR10e soldering cell – Cranfield, MATLAB, in progress **Nov – Dec 2026**

- Built the DH model and the forward and inverse kinematics of the UR3 in MATLAB and checked them, before the singularity analysis, path planning and control of the cell.

Sumo robot – ECAM, team of 3, 127-euro budget **Sep 2023 – Jan 2024**

- Main programmer of the Arduino logic (infrared edge sensors, three ultrasonic sensors) and customer contact.

EXPERIENCE

Engineering Intern – Réseaux de Chaleur Urbains d'Alsace, Strasbourg **Jun 2026 – Sep 2026**

District-heating operator running 30 networks. 12 weeks in the 15-person design office.

- Calculated the design basis of a closed-loop cooling system for a plant serving 40,000 dwelling-equivalents, replacing an open loop banned by environmental regulations.
- Showed the chiller had to be sized on summer rather than winter load, avoiding about 37 kW of unnecessary capacity.
- Delivered ten documents, including four tender documents and the hydraulic schematics drawn in BricsCAD.

Site Intern – Grands Travaux de l'Océan Indien (Colas Group), La Réunion **Jun 2025 – Aug 2025**

- Laid utility networks on a refrigerated warehouse site shared by about ten contractors, and spent a week alongside the site manager coordinating them.

EDUCATION

MSc Robotics – Cranfield University, UK **Sep 2026 – Sep 2027**

Robotics control, robotic perception, computer vision, machine learning, AI for robotics, ROS 2 and Gazebo, human-robot interaction, autonomy. Group project in digital robotics from March 2027.

Diplôme d'Ingénieur, ECAM Arts et Métiers – Strasbourg, France **Sep 2022 – Sep 2027**

Master's level. General engineering school built around five fields: industrial, mechanical, energy, electrical and electronic, and computer engineering, from CAD to supply chain and lean manufacturing, with many industrial partners that get students used to the corporate world early on. Erasmus semester at MCAST, Malta, in 2024.

SKILLS & INTERESTS

- **Languages:** French (native), English (C1, TOEIC 950/990), German (A2), Spanish (A2)
- **Programming:** C++ (object-oriented, STL), Python (OpenCV, Ultralytics YOLO, NumPy), Git and Linux, all currently being strengthened during the MSc
- **AI tools:** AI-assisted development with Claude Code, framed by written project rules
- **Robotics & embedded:** ABB RobotStudio, ROS 2 (introductory), Raspberry Pi, Arduino
- **Design:** SolidWorks, Fusion, Inventor, BricsCAD, EasyEDA, 3D printing (Prusa, PrusaSlicer), MATLAB (basic)
- **Interests:** volleyball in a university club, tennis, gym, cycling, photography and video editing